MARKET ANALYSIS

AUTONOMOUS RIDE-HAILING

Scenario-Based Opportunity Assessment – South Korea

September 2026

Jeonghyeon Lee

Full code, raw data, and chart visualization can be found in github.com/junghyori/Autonomous-Ride-Hailing-Korea-Market-Analysis

OVERVIEW

Where This Analysis Goes

01	**Purpose**	The problem this service would solve, and for whom, phase by phase.
02	**Opportunity**	Sizing the Korea market with a TAM → SAM → SOM framework.
03	**Go-to-Market**	A three-phase rollout sequenced by real demand and seasonality data.
04	**Price Logic**	A fare model that moves from adoption to monetization.
05	**Open Questions**	What isn't proven yet, and what we'd need to learn next.

Appendix – Data Sources & Methodology, Grid-Code Mapping, and Backup Charts

01 · PURPOSE

The Problem, and Who Feels It First

Problem: Korea's taxi industry faces a rapidly aging driver base and a well-documented late-night supply crunch, concentrated in a capital region of roughly 26 million people.

Solution: Autonomous ride-hailing adds scalable, technology-driven capacity — deployed on high-demand, well-mapped routes — that fills the gap without depending on new human drivers.

WHY - DATA POINTS

75%

of Seoul's 70K taxi drivers are aged 60 or older

Seoul (Private) Taxi Association data, 2025, via Asia Business Daily

~26M

people live across Seoul + Gyeonggi + Incheon (the capital area)

Statistics Korea (KOSIS), 2025 regional population estimate

~14K

Overall taxi complaints hit 14K in 2025 – the highest since 2022

Seoul Metropolitan Government, Transportation Complaint Statistics

FOR WHOM – BY PHASE

Phase 1

Tech-savvy professionals, corporate travelers, and airport shuttle riders – predictable routes, controlled environments.

Phase 2

General urban commuters in core Seoul districts – Gangnam, Hongdae, Yeouido.

Phase 3

Mass-market commuters across the full capital corridor – Seoul ↔️ Gyeonggi/Incheon.

Market Sizing: TAM → SAM → SOM

Grounded in Python-based analysis[(1)] of 2025 public taxi operations data from Korea's Ministry of Land, Infrastructure and Transport (국토교통부).
Rides scale down from national volume to a realistic Year-1 pilot.

National Taxi Trips
1,053M Trips/yr

All of Korea, 2025

TAM – Capital Area Trips
304M Trips/yr (29% of National)

TAM = Total Addressable Market.
Seoul + Gyeonggi + Incheon Annual Taxi Trips
(Data-Derived)

SAM – AV Suitable Trips
30.4M – 106.5M Trips/yr (10–35%)

SAM = Serviceable Addressable Market.
TAM × Addressable Share for AV Suitable Trips
(Assumption Bracket – see Open Questions)

SOM – Year 1 (Pilot Rides)
270K – 3.3M Rides

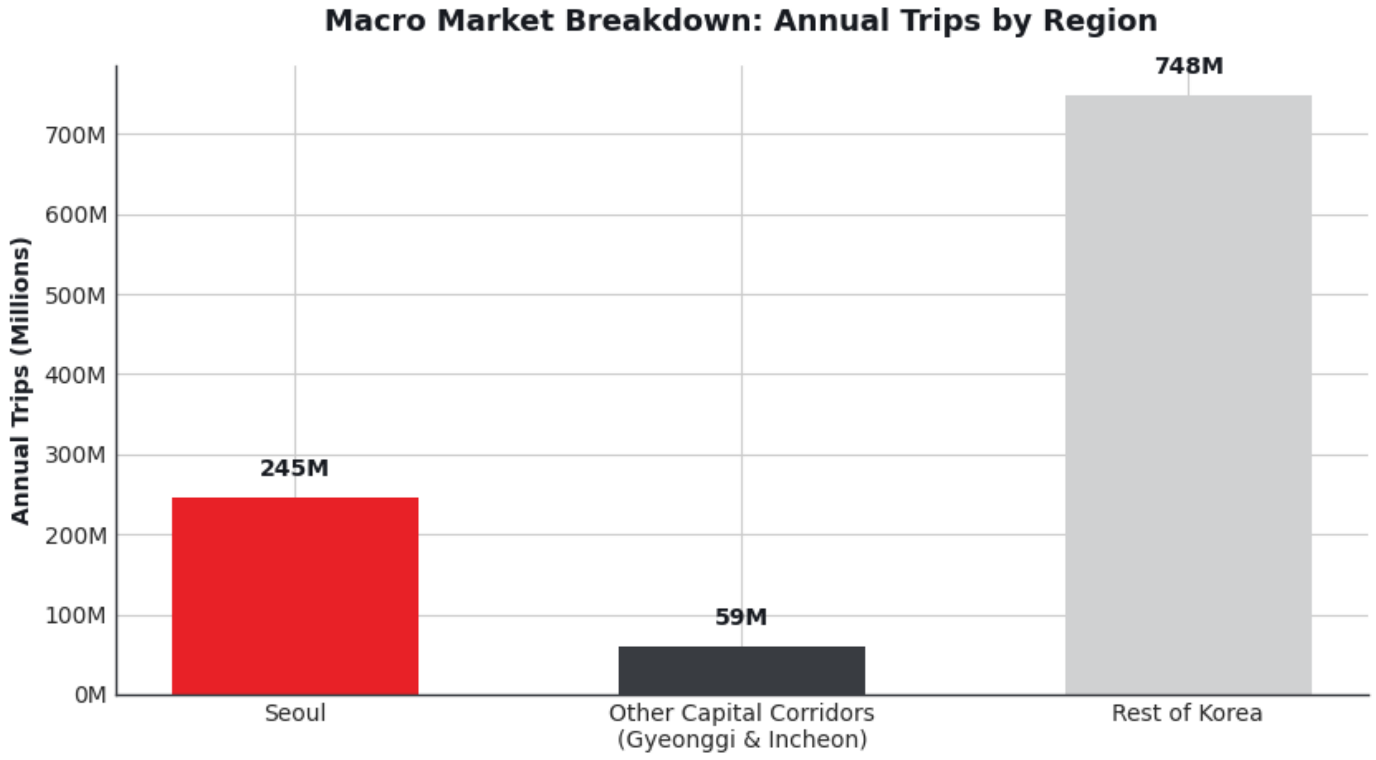
SOM = Serviceable Obtainable Market.
Pilot Fleet × Trips/Vehicle/Day × Operating Days
(Assumption Bracket – see Open Questions)

Source: *MOLIT TIMS, annual taxi-operating summary by municipality, 2025, via data.go.kr (dataset 15065131).*

(1) Full code, raw data, and every chart visualization can be found in **github.com/junghyori/Autonomous-Ride-Hailing-Korea-Market-Analysis**

Where National Taxi Demand Actually Sits

Seoul alone accounts for roughly a quarter of national taxi volume;
Adding Gyeonggi and Incheon brings the full capital area to ~29% of the country's annual trips – the geography this analysis treats as Total Addressable Market



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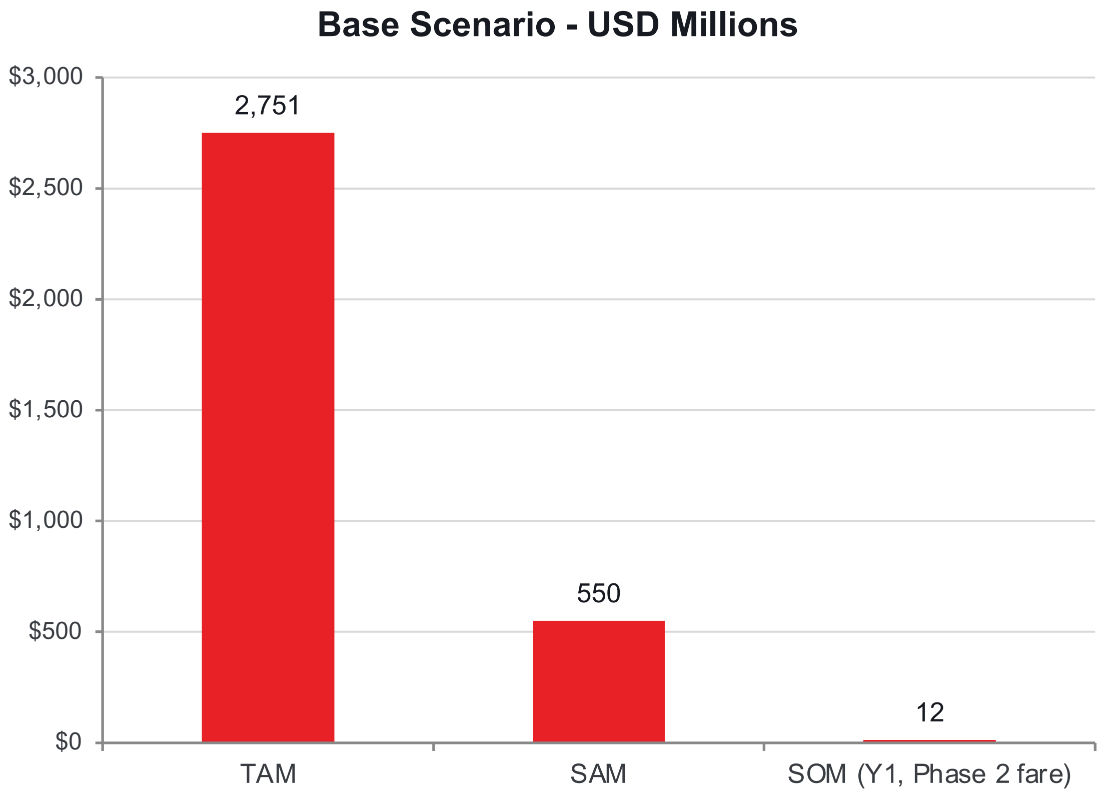
Revenue Potential Across Scenarios



Scenario	SAM Share	Y1 Pilot Fleet	Y1 Rides (SOM)	Y1 Revenue (USD)
Low	10%	100 vehicles	270K	$2.0–2.4M
Base	20%	300 vehicles	1.4M	$10.2–12.2M
High	35%	500 vehicles	3.3M	$24.8–29.8M

SAM (AV Suitable Market) Revenue Potential:
$275M (Low) – $550M (Base) – $963M (High)
at the realistic metered fare (≈KRW 12,200 / trip).

- Revenue range reflects Phase 1 (KRW10,164 metered) → Phase 2 (KRW12,202 metered) fares
- SOM assumes 10–20 trips/vehicle/day over 270–330 operating days

METHODOLOGY

Revenue (KRW) = Trips × Metered Fare/trip (₩10,164 Phase 1 / ₩12,202 Phase 2), based on ₩4,000 / ₩4,800 base fares (Phase 1 / Phase 2). USD = KRW ÷ ~1,350

Sources: *MOLIT TIMS, annual taxi-operating summary by municipality, 2025, via data.go.kr (dataset 15065131); Seoul Metropolitan Government official fare notice, See Slide 14 for detailed base-fare logic.*

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3-Phase Rollout Plan

Each phase targets a specific user segment, corridor, and fleet scale — grounded in actual 2025 hub-level trip volumes and seasonality, building operational confidence before broadening. Demand evidence and detailed plan by each phase can be found from next slide.

Phase 1

Months 0–6

Controlled Launch

ZONES

Pangyo Station + Incheon Airport T1/T2

FLEET

100 vehicles

FARE

KRW 10,164 / trip (-16.7% of metered)

Predictable,
low-complexity routes with tolerant early adopters.

Phase 2

Months 6–12

Urban Expansion

ZONES

Gangnam + Hongdae + Yeouido

FLEET

300 vehicles

FARE

KRW 12,202 / trip + 1.1–1.3x surge

Highest-density Seoul corridors,
tests nightlife and B2B commuter use cases.

Phase 3

Months 12+

Full Commercialization

ZONES

Full capital corridor (Seoul-Gyeonggi/Incheon)

FLEET

500+ vehicles

FARE

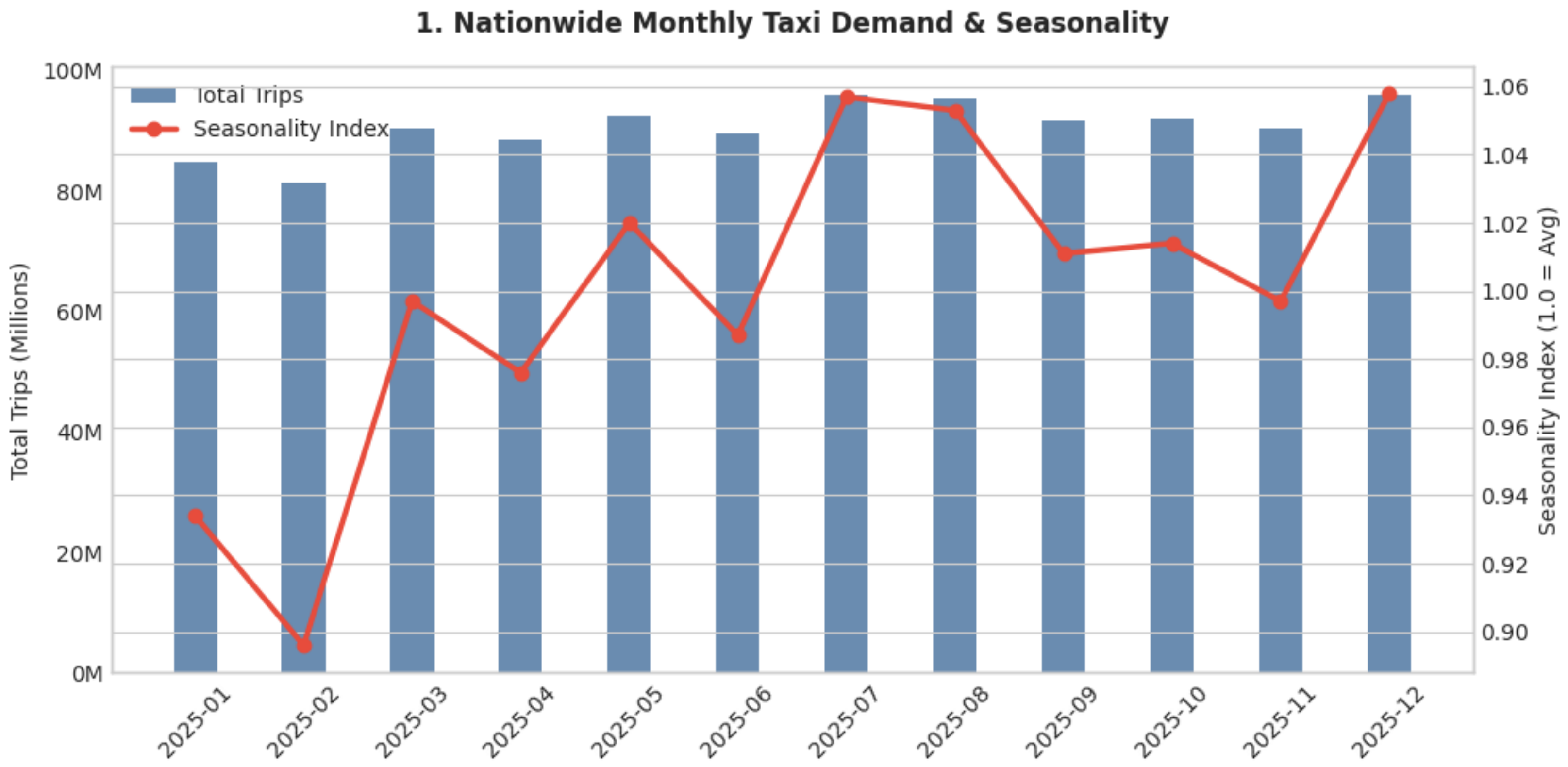
Subscriptions & Passes + KRW 12,202

Mass-market scale-up with retention-oriented pricing.

See Slide 14 for detailed base-fare logic.

Nationwide Demand Shows Seasonal Rhythm

National taxi trip volume dips in February and rises through mid-year, peaking around the summer and again at year-end. This rhythm sets the macro timing behind the rollout — e.g., the December peak used to maximize Phase 3 fleet utilization.

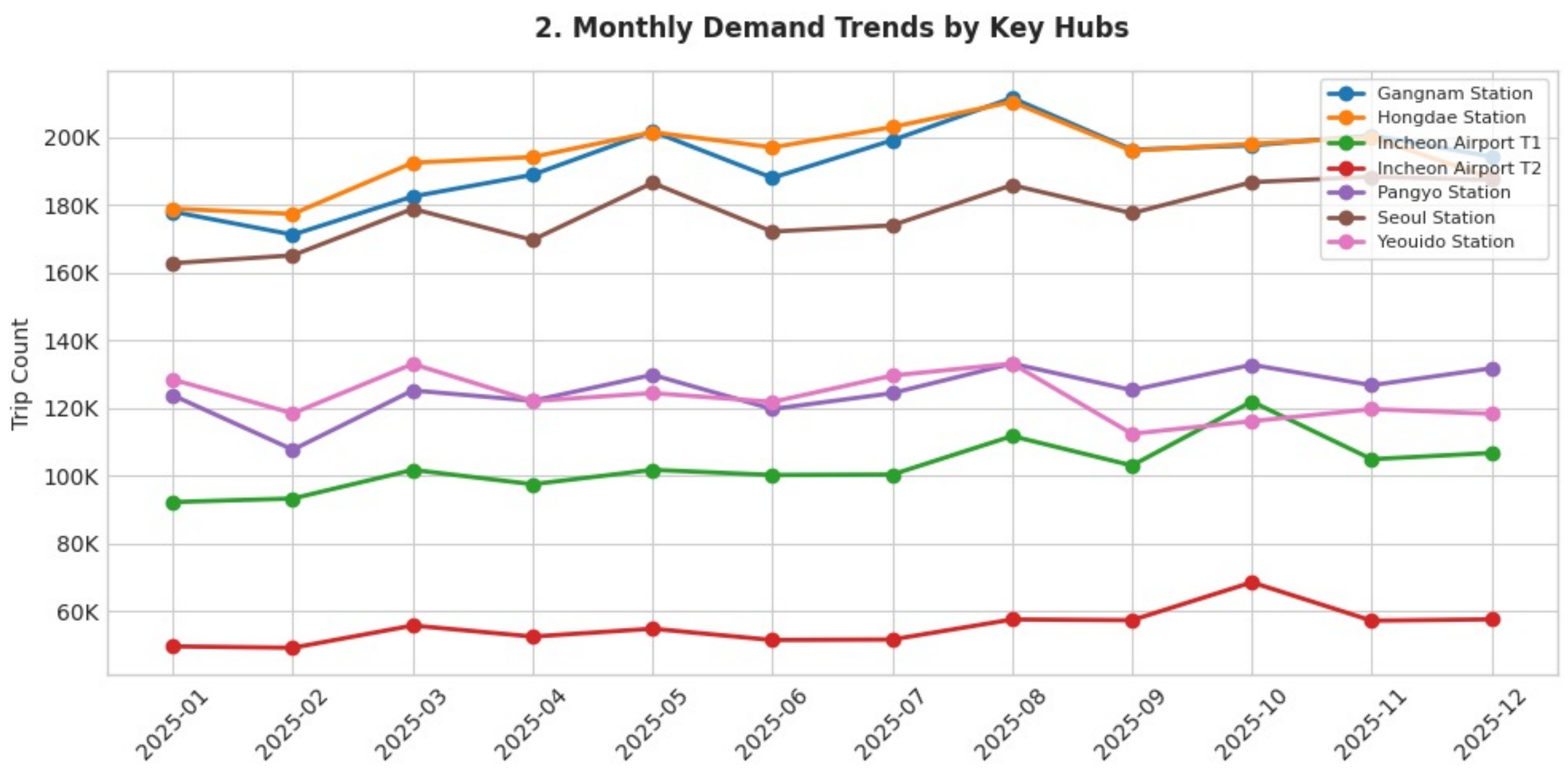


Source: *MOLIT TIMS, 500m-grid monthly taxi operating statistics, 2025, via data.go.kr (dataset 15160181).*
Seasonality index = month / annual monthly average.

Full code, raw data, and every chart visualization can be found in **github.com/junghyori/Autonomous-Ride-Hailing-Korea-Market-Analysis**

Hub-Level Volumes Set the Rollout Order

Gangnam and Hongdae carry the highest absolute trip volumes of the seven hubs studied — the core reason they anchor Phase 2. Pangyo and the airport terminals run at a fraction of that volume, matching their role as a lower-risk Phase 1 proving ground.

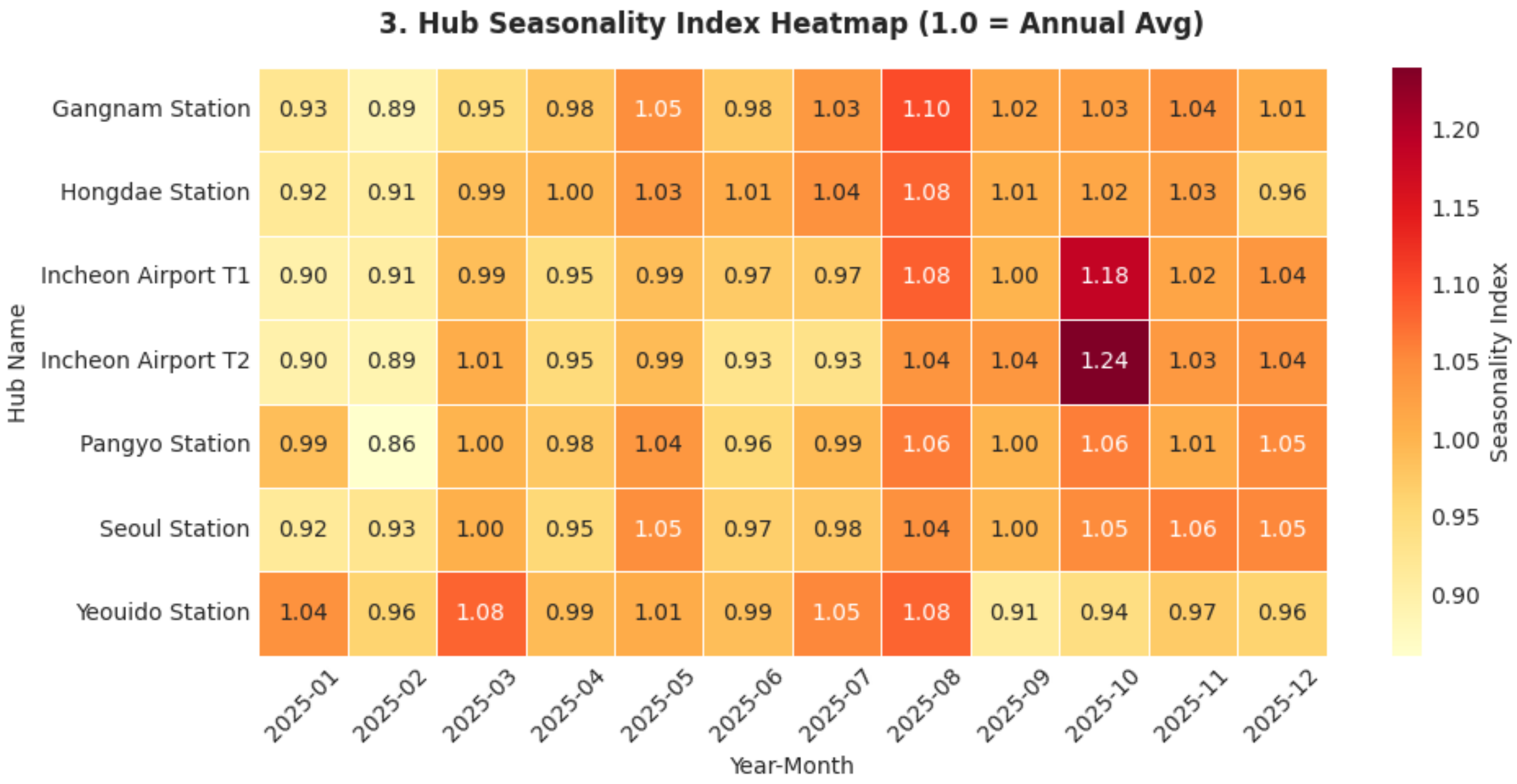


Source: *MOLIT TIMS, 500m-grid monthly taxi operating statistics, 2025, via data.go.kr (dataset 15160181).*
Hub grid codes mapped per the Korea National Grid System — see Appendix.

Full code, raw data, and every chart visualization can be found in **github.com/junghyori/Autonomous-Ride-Hailing-Korea-Market-Analysis**

Seasonality Differs Hub by Hub

Incheon Airport T2 spikes to 1.24x in October (Chuseok travel), while Yeouido dips to 0.91x in September.
These hub-specific swings — not just the national average — drive the seasonal fleet-redeployment plan in Phase 3.



Source: *MOLIT TIMS, 500m-grid monthly taxi operating statistics, 2025, via data.go.kr (dataset 15160181).*
Index of 1.00 = that hub's own annual monthly average.

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Controlled Launch (Months 0–6)

Zones: Pangyo Station + Incheon Airport T1/T2 · Fleet: 100 vehicles · Fare: KRW 10,164 (-16.7% vs. KRW 12,202 metered baseline)

Pangyo Station

125K trips / month

Seasonality: Stable year-round (0.86–1.06 seasonality index)

PROS

Tech-savvy riders (IT cluster), predictable commute patterns, low traffic complexity.

CONS

Low absolute volume caps early revenue upside.

WHY FIRST

Low-risk proving ground for AV technology with tolerant early adopters.

Incheon Airport T1 / T2

T1: 103K · T2: 55K trips / month

Seasonality: Sharp Chuseok peak – T2 hits 1.24x in October; winter dip Jan–Feb (0.89x)

PROS

Fixed origin-destination (terminal to city center), controlled roads, high fare per long-distance trip.

CONS

Seasonal spikes require flexible fleet sizing.

WHY FIRST

High-value corridor with simple routing; international travelers already have AV exposure.

Seasonality play: launch in March–April, as demand rises from the February trough, to build operational confidence before the summer peak.

Full code, raw data, and every chart visualization can be found in **github.com/junghyori/Autonomous-Ride-Hailing-Korea-Market-Analysis**

Urban Expansion (Months 6–12)

Zones: Gangnam + Hongdae + Yeouido · Fleet: 300 vehicles · Fare: KRW 12,202 + 1.1–1.3x mild surge

Gangnam Station

193K trips/mo — highest

Seasonality: Peak in August (1.10x)

PROS

Highest demand density, premium rider mix (business / entertainment).

CONS

Complex traffic, narrow streets, high pedestrian density.

Hongdae Station

195K trips/mo

Seasonality: Peak in August (1.08x)

PROS

Comparable volume to Gangnam; strong nightlife demand.

CONS

Late-night demand needs extended hours — safety/regulatory risk.

Yeouido Station

123K trips/mo

Seasonality: Volatile — Mar 1.08x, Sep 0.91x

PROS

Business district with regular weekday commute demand.

CONS

Sharp September dip tied to vacation season; lower weekend demand.

Seasonality play: time Phase 2 launch for May–June to capture the Jul–Aug summer peak (1.03–1.10 index) across all three hubs.
Volume across these hubs justifies fleet expansion once the tech is proven in Phase 1.

Full code, raw data, and every chart visualization can be found in **github.com/junghyori/Autonomous-Ride-Hailing-Korea-Market-Analysis**

Full Commercialization (Months 12+)

Zones: Full capital corridor (Seoul/Gyeonggi/Incheon) · Fleet: 500+ vehicles · Model: Subscriptions & Passes / Fare: KRW 12,202 + 1.1–1.3x mild surge

Seoul Station

178K trips / month · steady (0.92–1.06 seasonality)

Major transit interchange connecting KTX and subway — steady demand, But congested pickup zones require multi-modal coordination.

Cross-Corridor Expansion

Gyeonggi / Incheon: 59.4M trips / year

Massive untapped volume with simpler suburban road conditions, offset by longer trip distances (higher per-trip cost) and sparse off-peak demand.

RETENTION & FLEET BALANCING

Retention: Monthly commuting passes (Pangyo ↔ Gangnam, Airport ↔ Seoul), fixed-route shuttles, and tiered/premium options.
Seasonal balancing: Use the October airport surge (T2: 1.24x) and December national peak (1.06x) to maximize utilization; redeploy vehicles from Yeouido (Sep dip: 0.91x) to airport routes during Chuseok.

3-Phase Fare Progression

Anchored to a realistic metered fare (KRW 12,202/trip) — derived from actual Seoul trip data using the standard fare structure, not the KRW 4,800 base rate alone — pricing runs from penetration to monetization.

Phase 1 — Market Entry

0–6 MONTHS

KRW 10,164

-16.7% vs. metered fare

SURGE / MODEL

Flat, no surge

Lower adoption barriers;
secure initial trial volume with upfront flat quotes that remove surcharge anxiety.

Phase 2 — Stabilization

6–12 MONTHS

KRW 12,202

Full metered fare

SURGE / MODEL

Mild 1.1x–1.3x at peak / bad weather

Reach fare parity with standard taxis;
optimize fleet efficiency during peak windows.

Phase 3 — Expansion & Retention

12+ MONTHS

Sub + KRW 12,202

Passes & tiers

SURGE / MODEL

Bundled / Tiered
Mild 1.1x–1.3x at peak / bad weather

Maximize lifetime value with monthly passes, fixed-route shuttle packages, and premium options for frequent riders.

METHODOLOGY

Phase 1 discount = 1 - (₩4,000 ÷ ₩4,800) = 16.7%, applied to the metered fare: ₩12,202 × (1-16.7%) = ₩10,164
Fare Structure : ₩4,800 base (first 1.6km) + ₩100/131m distance + ₩100/30sec time (below 15.72km/h) + night surcharge (22:00–04:00: +20–40%) → avg trip (8.8km, 19min) blends to ₩12,202

Source: *Seoul Metropolitan Government official fare notice, news.seoul.go.kr/traffic/archives/1659.*

Full code, raw data, and every chart visualization can be found in **github.com/junghyori/Autonomous-Ride-Hailing-Korea-Market-Analysis**

What We Would Not Claim Yet

Regulatory & Legal

- Realistic timeline for Level 4 AV permits on Seoul municipal roads
- How insurance and liability would be structured, and at what premium

Consumer & Demand

- Actual willingness to ride driverless — stated vs. revealed preference gaps are large
- Effect of monsoon season and winter ice on uptime and rider acceptance

Competitive Landscape

- How Kakao Mobility, Tmap, and taxi cooperatives would respond
- Whether other AV operators are targeting the same corridors

Operational & Financial

- Realistic per-vehicle cost (CAPEX + maintenance + monitoring) vs. driver labor
- Achievable Year-1 utilization — our 10–20 trips/day range is assumed, not measured
- Fleet scale (100–500 vehicles by Month 12) is faster than how Waymo scaled: ~3 years post-launch (Phoenix, 2020) to reach ~700 vehicles across two metro areas

Data Limitations

- 2025 data reflects human-driven patterns; AV routing/demand may differ
- Addressable share (10–35%) is an assumption bracket, not a regulatory assessment
- MOLIT TIMS trip counts haven't been cross-validated against other published ridership benchmarks — treat absolute levels as directional

Labor & Union Opposition

- Korean taxi driver unions have organized large-scale protests against ride-hailing entrants before (e.g., Tada, 2019-2020)
- No union response, stakeholder engagement, or driver-transition plan is modeled in this analysis
- Regulatory timeline could shift materially if organized opposition escalates

Data Sources & Methodology

A. Primary Dataset – Taxi Operating Statistics

Ministry of Land, Infrastructure and Transport (MOLIT, 국토교통부) – Taxi Operation Information Management System (TIMS), via Korea's Public Data Portal.
Two datasets are used:
Annual summary by municipality (dataset 15065131): **data.go.kr/data/15065131**
500m-grid monthly detail (dataset 15160181): **data.go.kr/data/15160181**

B. Taxi Fare Reference

Seoul standard taxi fare (₩4,800 base + distance/time/night surcharges), per an official Seoul Metropolitan Government notice. Revenue uses a derived metered fare (₩12,202/trip), not the ₩4,800 base rate alone: **news.seoul.go.kr/traffic/archives/1659**

C. Full Analysis Notebook – Every Figure, Formula & Code

This entire analysis – the raw data pull, every formula, every chart shown in this deck (and a few not shown), and the full Python code – is reproducible from the public Jupyter notebook: **github.com/junghyori/Autonomous-Ride-Hailing-Korea-Market-Analysis**

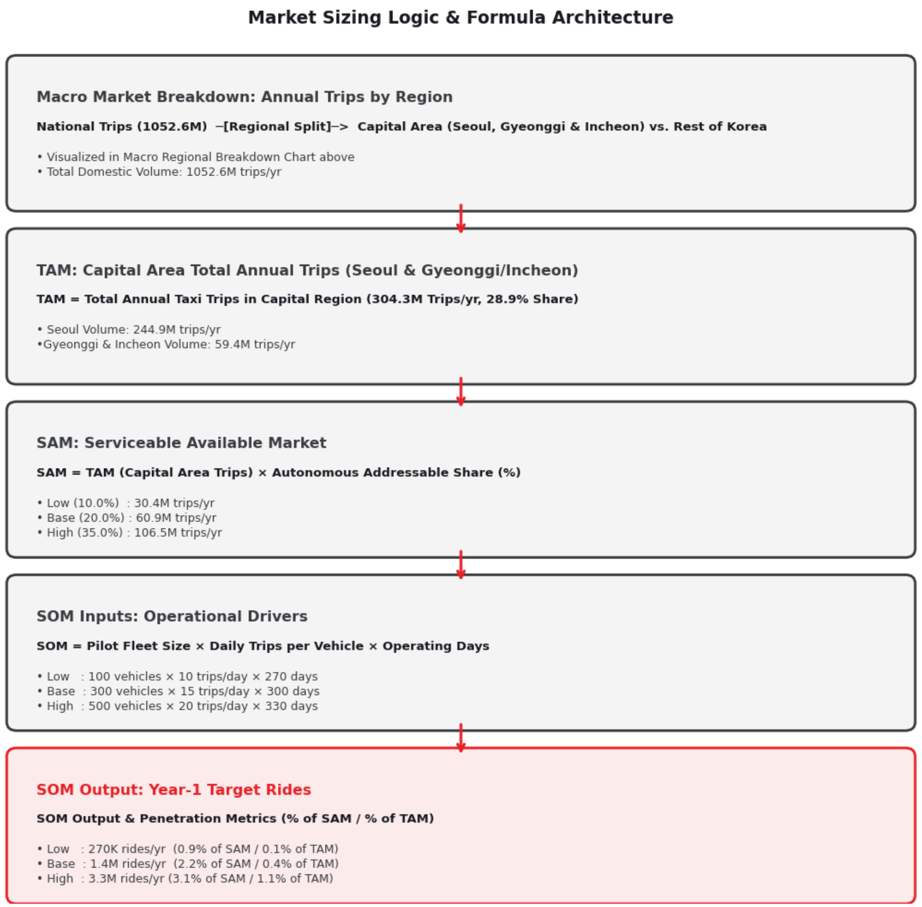
How Grid Codes Map to Real Locations

The Korea National Grid System (KNGS, 국가지점번호) divides all of South Korea into 100 km × 100 km cells, each labeled with a two-character Korean syllable code (e.g., **다사**, **마라**). Each 100 km cell is further divided into 500 m × 500 m sub-grids, indexed by a six-digit X/Y coordinate (e.g., **014078**). A full code such as **다사014078** therefore uniquely identifies one 500 m × 500 m cell in Korea — the level of detail used to define each demand hub in this analysis.

Grid code	Mapped location	Hub used in this analysis
다사014078	Incheon Int'l Airport Terminal 1 area (Yeongjong-do)	Incheon Airport T1
다사011083	Incheon Int'l Airport Terminal 2 area (Yeongjong-do)	Incheon Airport T2
마라080162	Gangnam / Sinnonhyeon Station area (Gangnam-gu, Seoul)	Gangnam Station
다사106101	Hongik Univ. / Hapjeong Station area (Mapo-gu, Seoul)	Hongdae Station
다사116089	Yeouido / National Assembly Station area (Yeongdeungpo-gu, Seoul)	Yeouido Station
마마003130	Seoul Station / Gwanghwamun area (Jung-gu, Seoul)	Seoul Station
마라078158	Seongnam, Gyeonggi-do (same 100km cell as Gangnam, SE of central Seoul)	Pangyo Station

Source: *National Geographic Information Institute (NGII) grid-lookup service, map.ngii.go.kr, and the V-World national spatial data portal, map.vworld.kr.*

Full TAM → SAM → SOM Calculation Detail



Source: *Reproduced directly from the analysis notebook; underlying figures from MOLIT TIMS, 2025, via data.go.kr.*

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THANK YOU

Autonomous Ride-Hailing – Korea Market Opportunity

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